A Finite-Volume Method for Predicting a Radiant Heat Transfer in Enclosures With Participating Media

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A new "finite-volume" method is proposed to predict radiant heat transfer in enclosures with participating media. The method can conceptually be applied with the same nonorthogonal computational grids used to compute fluid flow and convective heat transfer. A fairly general version of the method is derived, and details are illustrated by applying it to several simple benchmark problems. Test results indicate that good accuracy is obtained on coarse computational grids, and that solution errors diminish rapidly as the grid is refined.

Introduction

Fluid flow and convective heat transfer can now be predicted with reasonable economy and accuracy in very complex geometries. Most methods use finite difference, finite volume, or finite element methods to reduce the continuous equations of motion to a set of coupled algebraic equations that can be solved by computer. For many situations, e.g., flows with combustion, it would be desirable to add a solution for radiant exchange to the code that solves the fluid flow and internal energy equations; it would be convenient if the radiation model were based on the same philosophy and perhaps even used the same computational grid, as the fluid flow solver. Among the solution methods that are currently available, such as the many variations on the Monte Carlo, Zonal, Ray Tracing, Flux, and Discrete Ordinates methods none is fully satisfactory for this purpose except in the limit when the diffusion approximation applies (optically dense medium). Since excellent recent reviews of the state of the art are provided by Viskanta and Mengüç (1984, 1987), the competing methods for predicting radiative heat exchange will not be reviewed here.

According to the finite volume method, a solution domain is subdivided by a computational mesh that defines nodes for each variable and a control volume surrounding each node. To compute flows in complex domains, the flexibility allowed by a nonorthogonal boundary-fitted mesh is desirable. The present paper describes a new method for predicting radiative heat transfer that also uses such a grid. Besides using the same mesh as the flow solver, the method accounts for absorption, emission, and scattering in the medium, permits spectral dependence through a broad-band model (Viskanta and Mengüç, 1987), allows radiation properties to depend on other variables (like local pressure, concentration and temperature), and permits desired reflectivity properties of the wall to be incorporated. The method is also fully conservative in the sense that the global conservation balance for each discrete intensity component, as well as for the radiative flux, is exactly satisfied by the discrete equations.

The model has, until now, only been exercised for simple cases. These include heat transfer in rectangular enclosures, with diffuse gray isothermal walls, that contain gray emitting-absorbing and/or isotropically scattering media. To limit the model description to these cases would, however, make it difficult to appreciate the general applicability of the model. On the other hand, to provide the model in its full generality would result in notational complexity that would encumber the presentation. As a compromise, generality is restricted in those areas where removing the restriction is obvious and simple (e.g., gray medium and diffuse wall). The complexity of a nonisotropically scattering medium and a nonorthogonal grid is retained in the model development, even though these are not used in the test problems.

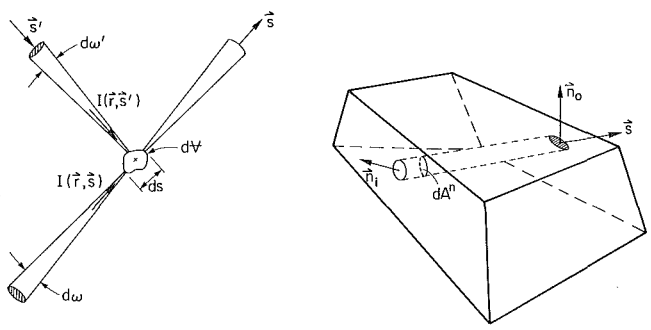
Finite Volume Approximation

The Volume-Integrated Equation for Intensity. Referring to the definition sketch (Fig. 1A) **r** denotes a location in space, $d\omega$ is a differential solid angle that is centered on the unit vector **s**, and s is distance along a particular pencil of radiation that lies in the **s** direction. Over the path length ds in the direction of the unit vector **s**, the rate of increase of total intensity $I(\mathbf{r}, \mathbf{s})$ is (Siegel, 1981)

$$\frac{dI}{ds} = -(K+\sigma^s)I + (KI_b + \sigma^s \bar{I}) \tag{1}$$

where the first term on the right-hand side is attenuation through absorption and outscattering and the second term is augmentation due to emission and inscattering. Because of the gray-medium approximation, the absorption coefficient K and the scattering coefficient σ^s are appropriate wavelength-averaged quantities. The inscattering term is given by

$$\bar{I}(\mathbf{r}, \mathbf{s}) = \frac{1}{4\pi}\int_{4\pi} I(\mathbf{r}, \mathbf{s}')\Phi(\mathbf{s}, \mathbf{s}')d\omega' \tag{2}$$



A B

Fig. 1 Part A shows notation used in the model. The control volume, for which the integral equation for intensity is derived, appears in B.

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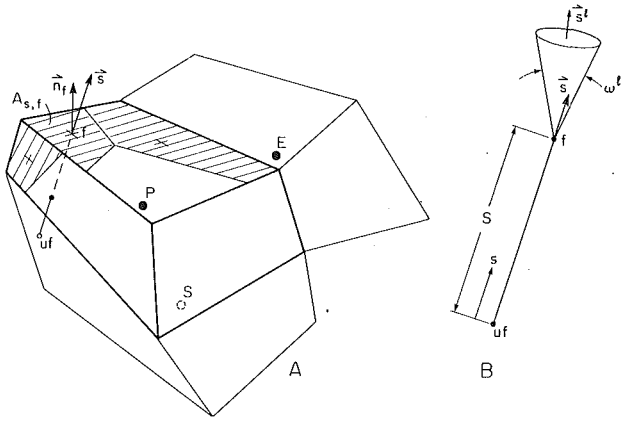


Fig. 2 Discrete surface panels on the discrete volume centered at *P*, and neighbor control volumes. The intensity I^l_f, from which the flux through $A_{s,f}$ is found, is expressed as an "upstream" value I^l_{uf} found by interpolation from nodal values, and a change between *uf* and *f*.

where $I(\mathbf{r}, \mathbf{s}')$ is the intensity at $\mathbf{r}$ within the solid angle $d\omega'$ that lies in the direction of the unit vector $\mathbf{s}'$.

Integration of equation (1) over any specified volume yields the desired integral equation. Choosing the hexahedron in Fig. 1(B) for purposes of illustration, the volume is subdivided into pencils along $\mathbf{s}$ of the cross-sectional area dA^n. Multiplication of equation (1) by the differential volume $d\mathcal{V} = dA^n ds$, carrying out the integration, and multiplying the resulting equation by $d\omega$, yields

$$\int_{A_s} I_S(\mathbf{s}\cdot\mathbf{n})dA_S d\omega = \int_{\mathcal{V}} [-(K+\sigma^s)I + KI_b + \sigma^s \bar{I}]d\mathcal{V}d\omega \qquad (3)$$

where A_s and $\mathcal{V}$ are the surface area and volume of the hexahedron, $\mathbf{n}$ is the unit surface normal, and I_s is the intensity at the surface. This equation demands that the net radiant energy within $d\omega$ that leaves through all surfaces of the volume is exactly balanced by the excess, within the volume, of inscattering and emission over outscattering and absorption.

Discrete Equation, Level 1 Approximation. According to the finite volume method, space within the interior of the domain of interest is subdivided into discrete nonoverlapping volumes, and a single "node" is located centrally within each volume. The control volume in Fig. 1(B) is redrawn in Fig. 2, showing its central node P, and nodes E and S for the neighboring control volumes to the east and south.

Since direction is also an independent variable, it is consistent with the finite volume method to subdivide direction into N^l discrete, nonoverlapping, solid angles of size ω^l, $l = 1, 2, 3, \ldots$, which sum to 4π. The number of angles and their size distribution is, like the discretization of space, at the discretion of the analyst.

The objective of the finite volume method is to find I^l_P, the intensity at each node P associated with each discrete bundle ω^l. An algebraic equation for I^l_P, is obtained by introducing approximations into equation (3).

Choosing the volume in equation (3) to be the control volume associated with node P in Fig. 2, with surface area $A_{S,P}$, and integrating the equation over the discrete solid angle ω^l, (3) becomes

$$\int_{\omega^l}\int_{A_{S,P}} I_S(\mathbf{s}\cdot\mathbf{n})dA_S d\omega =$$

$$\int_{\omega^l}\int_{\mathcal{V}_P} [-(K+\sigma^s)I + KI_b + \sigma^s \bar{I}]d\mathcal{V}d\omega \qquad (4)$$

The simplest approximation of the right-hand side of equation (4) is

$$\int_{\omega^l}\int_{\mathcal{V}_P} [-(K+\sigma^s)I + KI_b + \sigma^s \bar{I}]d\mathcal{V}d\omega$$

$$\approx [-(K_P+\sigma^s_P)I^l_P + K_P I^l_{bP} + \sigma^s_P \bar{I}^l_P]\mathcal{V}_P\omega^l \qquad (5)$$

In carrying out the volume integrations, all variables have been assumed constant over $\mathcal{V}_P$ and ω_l. A more accurate representation, involving, for example some linear distribution between nodes, is possible but is both more complex and may have serious negative computational effects (like a strong coupling between intensities in different directions that is not consistent with equation (1)).

As the first step to approximating the left-hand side of equation (4), the surface of the control volume is subdivided into N_f surface panels of area $A_{S,f}$, a few of which are shown in Fig. 2(A). Denoting the radiant transfer through panel f and within ω^l by Q^l_f, the left-hand side of equation (4) is just the

Nomenclature

A_s = surface area of control volume, m^2
$A_{s,f}$ = surface area of panel f of control volume, m^2
B_P = source term in algebraic equation, e.g., equation (24)
D = depth of one-dimensional plane layer, m
$\mathbf{e}_j$ = unit vector in the j direction
I = radiation intensity, W/m^2Sr
$\bar{I}$ = scattered intensity, W/m^2Sr, see equation (2)
I_a = average intensity, equation (13)
I_b = blackbody intensity $= \sigma T^4/\pi$
I^l_B = discrete intensity at the bottom surface within ω^l
I^l_P = discrete intensity at node P within ω^l
I_S = intensity at surface control volume
I^l_T = discrete intensity at the top surface within ω^l
K = absorption coefficient, m^{-1}
L_x, L_y = dimensions of rectangular cavity in x and y directions, m
m = index denoting the ω^l with boundaries at θ^m_+ and θ^m_- (see Fig. 3(B))
n = index denoting the ω^l with boundaries at ϕ^n_+ and ϕ^n_-
$\mathbf{n}$ = unit vector normal to a control volume surface
N^l = total number of discrete solid angles
N_x, N_y, N_z = number of control volumes in the x, y, and z directions, respectively
N^θ, N^ϕ = number of discrete angles in θ and ϕ directions
q = heat flux, W/m^2
Q = total radiant heat transfer rate, W
Q^l_f = radiant heat transfer across surface panel f
$\mathbf{r}$ = position in space
R = $(KI_b + \sigma^s\bar{I})/\kappa$, W/m^2Sr
$\tilde{R}, \tilde{R}$ = see equation (21)

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sum of Q_f^l, $f=1, 2, \ldots, N_f$. The simplest approximation for Q_f^l is

$$Q_f^l = A_{S,f}\int_{\omega^l} I_f(\mathbf{s}\cdot\mathbf{n}_f)d\omega \tag{6}$$

where the intensity and unit normal to the surface have been approximated by their values at the integration point "f," located "centrally" on the panel.

The simplest Level 1 approximation of (3) is therefore

$$\sum_{f=1}^{N_f} Q_f^l = \sum_{f=1}^{N_f} A_{S,f}\int_{\omega^l} I_f(\mathbf{s}\cdot\mathbf{n}_f)d\omega$$
$$= [-(K_P+\sigma_P^s)I_P^l + K_P I_{b,P} + \sigma_P^s \bar{I}_P^l]\mathcal{V}_P\omega^l \tag{7}$$

Discrete Equation, Level 2 Approximation. To close the set of equations, relations are needed between the panel integration-point values I_f and nodal point values I_P, I_E, etc. Since several different closure relations, all starting from equation (7), are possible, these are defined as the Level 2 approximations.

In the present study, I_f was found by tracing back along the path taken by the ray in reaching f (i.e., along the -s direction) until a location was reached at which the intensity can be reasonably found by interpolation between nodes. The location of the interpolation point will be specified later when the grids are generated. For now, it is sufficient to denote its location by uf as shown in Fig. 2 and the intensity (known by the interpolation) by I_{uf}. To estimate the value of I_f from I_{uf} for the given s direction, rewrite equation (1) as

$$\frac{dI}{ds} + \kappa I = \kappa R \tag{8A}$$

where

$$\kappa = K+\sigma^s \text{ and } R = (KI_b + \sigma^s \bar{I})/\kappa \tag{8B}$$

Let R be approximated by the first two terms of a Taylor series expansion about point f, and treat κ as constant at κ_f. Carrying out the integration of equation (8A) over distance S from uf to f, shown in Fig. 2(B), the intensity in direction s at integration point f is

$$I_f = I_{uf}e^{-\kappa_f S} + R_f(1-e^{-\kappa_f S}) - \frac{(\partial R/\partial s)_f}{\kappa_f}(1-e^{-\kappa_f S}(1+\kappa_f S)) \tag{9}$$

$(\partial R/\partial s)_f$, the directional derivative in the s direction, is related to the gradient of R, ∇R, by

$$\left(\frac{\partial R}{\partial s}\right)_f = (\nabla R)_f\cdot\mathbf{s}$$

Introducing equation (9) into equation (6), and carrying out the integration over ω_l, gives the following estimate of the radiant heat transfer within ω^l that crosses surface panel f:

$$Q_f^l \approx A_{S,f} I_{uf}^l e^{-\kappa_f \bar{S}} N_f^l + A_{S,f} R_f^l (1-e^{-\kappa_f \bar{S}})N_f^l$$
$$-\frac{A_{S,f}}{\kappa_f}(1-e^{-\kappa_f\bar{S}}(1+\kappa_f\bar{S}))\sum_{j=1}^{3}\left(\frac{\partial R^l}{\partial x_j}\right)_f D_{jf}^l \tag{10}$$

where R^l is the value of R for the bundle ω^l, where the gradient of R has been arbitrarily written in Cartesian coordinates x_j, defined by the unit vectors $\mathbf{e}_j$, as

$$\nabla R^l = \sum_{j=1}^{3}\left(\frac{\partial R^l}{\partial x_j}\right)\mathbf{e}_j$$

and where N_f^l and D_{jf}^l are integrals that depend only on the panel orientation for a given ω^l

$$N_f^l = \int_{\omega^l}(\mathbf{s}\cdot\mathbf{n}_f)d\omega \qquad D_{jf}^l = \int_{\omega^l}(\mathbf{e}_j\cdot\mathbf{s})(\mathbf{s}\cdot\mathbf{n}_f)d\omega \tag{11}$$

In the integration over ω^l in equation (6) that led to equation (10), I_{uf} was treated as constant, and S was held constant at $\bar{S}$. Their values are evaluated along the ray $\mathbf{s}^l$ that lies in the center of the solid angle ω^l. To include the effect of varying S and I_{uf} over ω^l would dramatically increase complexity. Furthermore, the practice of using constant values is consistent with the other approximations embodied in the discrete equation representation of the continuous equations.

Before continuing, it is of interest to demonstrate that the approximations of the heat flux given by equation (10) is reasonable. For concreteness, consider the case where the surface panel lies in a z plane, with surface normal $\mathbf{e}_3$. For the optically thin limit ($\kappa_f \to 0$), the radiant flux becomes

$$Q_f^l \to A_{S,f} I_{uf}^l \int_{\omega^l}(\mathbf{s}\cdot\mathbf{e}_3)d\omega$$

The intensity in this limit is propagated correctly (i.e., without

Nomenclature (cont.)

s = distance along a beam in the s direction, m
$\mathbf{s}$ = unit vector in the direction for which I is to be calculated
$\mathbf{s}^l$ = unit vector in the center of discrete solid angle ω^l
S = distance from the interpolation point to the integration point (see Fig. 2(B))
$\mathbf{S}$ = average value of S over the solid angle ω^l
T = temperature, K
$\mathcal{V}$ = volume, m^3
x, y, z = Cartesian coordinates, m
x_1, x_2, x_3 = Cartesian coordinates; $x_1=x$, $x_2=y$, $x_3=z$, m
x^* = $(\kappa+K)D$, used for one-dimensional problem
ϵ = surface emissivity
θ = polar angle, measured from the z axis
κ = extinction coefficient $=K+\sigma^s$, m^{-1}
ρ = surface reflectivity
σ = Stefan–Boltzmann constant = 5.729×10^{-8} W/m^{2}K^4
σ^s = scattering coefficient, m^{-1}
ϕ = azimuthal angle in the x–y plane, measured from x axis (Fig. 6)
Φ = phase function for scattering from direction $\mathbf{s}'$ to direction $\mathbf{s}$
ω = solid angle
ω^l = solid angle of the discrete bundle centered at $\mathbf{s}^l$

Subscripts

B = bottom surface
P, N, E, S, W = labels for nodes within control volumes
f = at the integration point f
uf = at the interpolation point (Fig. 2(A))
T = top surface
j = coordinate index = 1, 2, 3

Superscripts

l = in the solid angle ω^l
$*$ = nondimensional

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attenuation) from the interpolation point *uf* to the integration point on the panel.

In the optically thick limit ($\kappa_f \to \infty$)

$$Q_f^l \to A_{S,f} R_f^l \int_{\omega^l} (\mathbf{s}\cdot\mathbf{e}_3)d\omega - \frac{A_{S,f}}{\kappa_f} \sum_{j=1}^{3} \left(\frac{\partial R^l}{\partial x_j}\right)_f \int_{\omega^l} (\mathbf{e}_j\cdot\mathbf{s})(\mathbf{s}\cdot\mathbf{e}_3)d\omega$$

Summing over l to obtain the net radiant heat transfer crossing the panel, and for isotropic scattering (so that R becomes independent of angle l)

$$Q_f = \sum_l Q_f^l = A_{S,f}\left(-\frac{4\pi}{3\kappa_f}\right)\left(\frac{\partial R}{\partial x_3}\right)_f$$

This is the correct diffusion approximation for the optically thick limit (Siegel, 1981). Thus the expression for Q_f^l given by equation (10) has retained sufficient generality to capture both the optically thin and thick limits. Note that, had the third term in equation (9) been omitted (which is the usual solution (Siegel, 1981) to (8*A*) for constant κ and R), the radiation heat transfer across the panel would not have recovered the diffusion approximation.

Returning to the development of the discrete equation for I_P^l, equation (10) is substituted into equation (7). The only remaining issues are how to interpolate I_{uf}^l from nodal-point values like I_P^l, and how to find R_f^l. As already mentioned, the interpolation will be illustrated when the method is implemented on a specific mesh. The value of R_f^l is obtained by a straightforward linear interpolation using values of R^l, such as R_P^l, that are calculated and stored at the nodes.

To find R_P^l at the nodes requires an evaluation of both its components (see equation (8*B*)) $\bar{I}_P^l$ and $I_{b,P}$. $\bar{I}_P^l$ is obtained from the following discrete approximations to equation (2):

$$\bar{I}_P^l = \frac{1}{4\pi} \sum_{l'} I_P^{l'} \Phi(l', l)\omega^{l'} \tag{12}$$

where $\Phi(l', l)$ is the phase function for scattering from the solid angle $\omega^{l'}$ to the solid angle ω^l. For isotropic scattering, $\Phi = 1$ and equation (12) becomes

$$\bar{I}_P^l = \frac{1}{4\pi} \sum_{l'} I_P^{l'}\omega^{l'} = I_{aP} \tag{13A}$$

where I_{aP} is the discrete approximation at node P of the average intensity I_a:

$$I_a(\mathbf{r}) = \frac{1}{4\pi}\int_{4\pi} I(\mathbf{r}, \mathbf{s}')d\omega' \tag{13B}$$

The value of the second component of R_p, i.e., $I_{bP} = \sigma T_P^4/\pi$, is found from the thermal energy balance for the P control volume. The increase in internal energy due to radiation crossing the boundaries of the control volume in Fig. 2(*A*) is obtained by replacing the integration limit ω^l in equation (4) by 4π.

$$-\int_{A_{S,P}}\int_{4\pi} I(\mathbf{s}\cdot\mathbf{n})d\omega dA_S = 4\pi\int_{\mathcal{V}_P} K(I_a - I_b)d\mathcal{V} \tag{14A}$$

$$\approx 4\pi K_P(I_{a,P} - I_{b,P})\mathcal{V}_P \tag{14B}$$

The scattering contribution drops out of this energy balance for both isotropic and nonisotropic scattering. The temperature is established, in general, by a complex interaction (Siegel, 1981) of this term with advection and diffusion of energy, and with work done by the pressure and other stresses. For the simplest case of radiative equilibrium, the fluid temperatures adjusts to make the net surface heat transfer in equation (14*A*) zero, so that

$$I_{b,P} = I_{a,P} \tag{15}$$

For the special case of isotropic scattering and radiative equilibrium, combining equations (8*B*), (13), and (15) results in

$$R_P^l = R_P = I_{a,P} = I_{b,P} \tag{16}$$

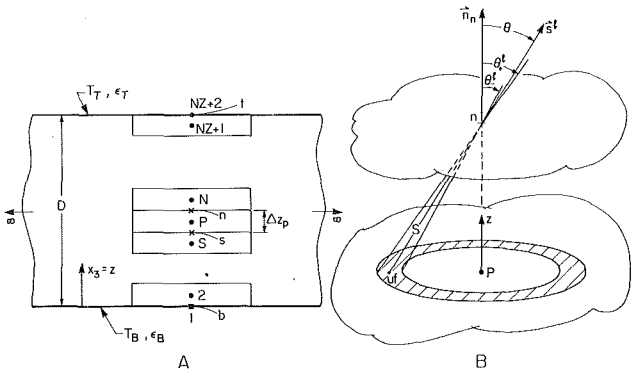


Fig. 3 Geometry and control volumes for radiation in a plane participating medium contained between isothermal gray walls

Summary and Discussion. This section has provided the details of a general finite volume method, except for the interpolation to obtain I_{uf}^l and the application of boundary conditions. These details, and the performance of the method, are illustrated in the following sections by application to some simple problems for which benchmark solutions are available.

One-Dimensional Radiant Heat Transfer in a Plane Medium

Problem Description. Very precise solutions (Heaslet, 1965) (see Viskanta and Mengüç 1987 for other references) have been obtained for radiant heat transfer between two isothermal, diffuse, flat plates, at temperatures T_B and T_T and emissivities ϵ_B and ϵ_T (see Fig. 3), that enclose a gray isotropically scattering or absorbing-emitting medium with constant scattering and absorption coefficients, and that is in radiative equilibrium. The finite volume method is first applied to this problem.

Discrete Equations. The natural control volumes for this problem are shown in Fig. 3(*A*). The depth Δz of each control volume is arbitrary and nodes are located midway between the control volume faces, except for the first and last nodes, which are located on the plates. Integration points, like n and s, which are denoted by $\times$, lie on the volume faces. On the walls, the integration points b and t lie in the medium distance δ from the walls, where $\delta \to 0$.

For this problem, the radiation intensity depends on z and θ (Fig. 3(B)). The discrete solid angle ω^l is defined as the annular cone between θ_-^l and θ_+^l. The interpolation point *uf* that is upstream of the integration point n is taken to lie on the plane of node P, as shown in the figure. From the assumption that the intensity varies spatially only with z, I_{uf}^l in equation (10) can be replaced by I_P^l for $\mathbf{s}\cdot\mathbf{e}_3 > 0$ and by I_N^l for $\mathbf{s}\cdot\mathbf{e}_3 < 0$. For $\mathbf{s}\cdot\mathbf{e}_3 > 0$, the radiant heat transfer within ω^l that travels through the control volume surface n that lies north of node P is, according to equation (10) and Fig. 3(*B*)

$$Q_n^l = F_n^l I_P^l + G_n^l R_n - H_n^l\left(\frac{\partial R}{\partial z}\right)_n \tag{17}$$

where the angular dependence of R has been dropped due to the isotropic scattering approximation. In this equation

$$F_n^l = A_S e^{-\kappa S^l} N^l \qquad G_n^l = A_S(1 - e^{-\kappa S^l})N^l \tag{18A}$$

$$H_n^l = \frac{A_S}{\kappa}(1 - e^{-\kappa S^l}(1 + \kappa S^l))D_3^l \tag{18B}$$

$S^l = (z_n - z_P)/\cos\theta^l$ is defined here as the distance S in Fig.

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3(B) for the angle $\theta = \theta^l = (\theta^l_- + \theta^l_+)/2$. From equation (11) for this geometry

$$N^l = \int_{\omega^l} (\mathbf{s}\cdot\mathbf{e}_3)d\omega = -\pi[\cos^2\theta^l_+ - \cos^2\theta^l_-] \qquad (18C)$$

$$D^l_3 = \int_{\omega^l} (\mathbf{s}\cdot\mathbf{e}_3)^2 d\omega = -\frac{2\pi}{3}[\cos^3\theta^l_+ - \cos^3\theta^l_-] \qquad (18D)$$

and the solid angle is

$$\omega^l = \int_{\theta^l_-}^{\theta^l_+} 2\pi \sin\theta d\theta = -2\pi[\cos\theta^l_+ - \cos\theta^l_-] \qquad (18E)$$

Substitution of equation (17) for Q^l_n, and a similar equation for the heat transfer through the south face Q^l_s, into equation (7), and collection of all terms involving R into a "source term," b^l_P, results in the following equation for I^l_P (for $\mathbf{s}\cdot\mathbf{e}_3 > 0$):

$$a^l_P I^l_P = a^l_S I^l_S + a^l_N I^l_N + b^l_P \qquad (19A)$$

where

$$a^l_P = F^l_n + \kappa \forall_P \omega^l \qquad a^l_S = F^l_s \qquad a^l_N = 0 \qquad (19B)$$

$$b^l_P = \left[G^l_s R_s - H^l_s\left(\frac{\partial R}{\partial z}\right)_s\right] - \left[G^l_n R_n - H^l_n\left(\frac{\partial R}{\partial z}\right)_n\right] + \kappa R_P \forall_P \omega^l \qquad (19C)$$

The values of R and $(\partial R/\partial z)$ at an integration point (i.e., at a face) are found by linear interpolation of the nodal point values or, at the top and bottom plates, by linear extrapolation from interior nodal values.

For a diffuse top surface, at temperature T_T, the boundary condition for the surface intensity I^l_T for radiation into the medium, $\mathbf{s}^l\cdot\mathbf{e}_3 < 0$, is

$$(1-\epsilon_T)\sum_{\mathbf{s}^{l'}\cdot\mathbf{e}_3>0} Q^{l'}_t + \epsilon_T A_S \sigma T^4_T = \sum_{\mathbf{s}^l\cdot\mathbf{e}_3<0} Q^l_T = A_S \pi I^l_T \qquad (20)$$

The terms on the left-hand side of equation (20) represent reflected and emitted radiation from the T surface, which constitute the total radiation into the medium. The last equality defines the intensity through the gray-surface approximation. A similar equation at the bottom surface relates the intensities into the medium ($\mathbf{s}^l\cdot\mathbf{e}_3 > 0$) to the bottom surface temperature and to intensities incident on the bottom.

Solution Method. For fixed nodal values of R and upward intensities at the bottom surface, I^l_P values are found for a given l by application of equation (19A) to each interior node, starting at the bottom and sweeping upward. This is repeated for each l for which $\mathbf{s}^l\cdot\mathbf{e}_3 > 0$. Equation (17) provides the incident fluxes onto the top surface, from which equation (20) gives the downward intensities. Equations similar to equation (19A), but with a^l_S rather than a^l_N equal to zero, are then solved for the downward intensities and incident fluxes on the lower surface. The lower surface boundary condition yields the upward boundary intensities to complete one cycle.

If the nodal values of R are fixed, one cycle yields the desired result if $\epsilon_B = \epsilon_T = 1.0$. Several cycles will be required for reflective boundaries.

The determination of R is, of course, part of the solution procedure. The simplest "explicit" method is to obtain the intensities for an assumed R in the manner just described, to substitute these into equation (13A) and thus obtain new R values from equation (16). This cycle is repeated until convergence is achieved. Convergence was deemed satisfactory when the change, between successive updates of R, in net heat flux at both surfaces was less than 0.001 percent.

Results. When a fine grid (see below) was used, the non-dimensional heat transfer, $q^* = q/\sigma(T^4_B - T^4_T)$, obtained using the finite volume method agreed to within 1 in the fourth significant figure with the predictions reported by Heaslet and Warming (1965). The gas temperature, and temperature jump at the boundary, were also predicted correctly.

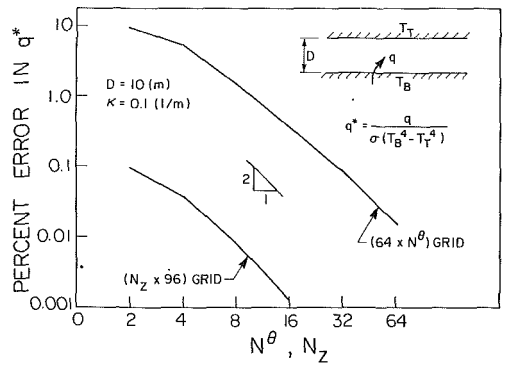


Fig. 4 Results of computational experiments to determine the rate at which the error in heat transfer from the surface decreases with grid refinement. The slope of 2:1 on the figure denotes quadratic convergence.

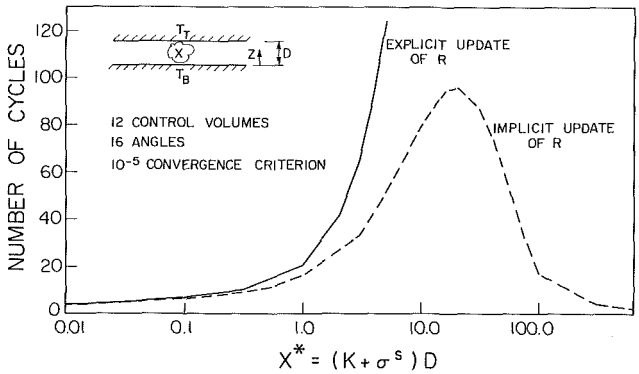


Fig. 5 The number of R-update cycles required to obtain a wall heat transfer that changed less than 0.001 percent between cycles

Table 1 Comparison of finite volume results for uniform grid of four control volumes and eight angles with "exact" results (Heaslet and Warming, 1965); $\kappa^* = \kappa D$

	$q^* = q/\sigma(T^4_B - T^4_T)$			
κ^*	Coarse grid finite volume results	Exact results	Percent error	CPU, s
0.0	1.0000	1.0000	0.0	0.1
0.1	0.9125	0.9157	0.3	0.2
0.5	0.6964	0.7040	1.1	0.3
1.0	0.5469	0.5532	1.1	0.4
2.0	0.3865	0.3900	0.9	0.4

Even for a coarse grid (see Fig. 3) of four uniform control volumes in the z direction and with polar angle θ divided into eight equal intervals, the finite volume method predicts the correct heat transfer to within about 1 percent, for a wide range of optical depths, as shown in Table 1. The CPU time required was less than 0.5 s on a MicroVax II computer.

Having established that the finite volume method does yield the correct results as the grid is refined, it is of major interest to establish how quickly the accuracy improves with grid refinement. For $\kappa^* = \kappa D$ of 1.0, the heat transfer was obtained using $N_z = 64$ uniform control volumes and $N^l = N^\theta = 96$ equal angle intervals. To study error reduction with N_z, q^* was recalculated on a sequence of grids of size $(N_z \times 96)$, $N_z = 2$, 4, . . . , 64. The results in Fig. 4 indicate the q^* is obtained to within 0.1 percent with $N_z = 2$, and for $N_z \geq 4$ the error becomes proportional to $(N_z)^{-2}$ (i.e., quadratic convergence).

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To study error reduction with angular refinement, q^* was calculated for grids of $(64\times N^\theta)$, $N^\theta = 2, 4, 6, \ldots, 96$. The error decreases from about 10 percent for $N^\theta = 2$, and for $N^\theta \geq 4$, and decreases roughly as $(N^\theta)^{-2}$.

Another issue of major interest is the number of updates of R required to obtain convergence. For black surfaces, and for a change of q^* at the surfaces of less than 0.001 percent between R updates, the number of iterations required by the explicit method is plotted against $\kappa^* = \kappa D$ in Fig. 5. As $\kappa^* \to 0$, the correct solution is obtained without iteration because the intensities become independent of R. The number rises sharply with κ^* but convergence is satisfactory for $\kappa^* < 4.0$, which includes the range of interest for many practical applications. The number of iterations increases sharply with κ^* for $\kappa^* \leq 4.0$. A similar behavior, of increasingly difficult convergence with increasing κ^*, has been encountered in the use of the discrete ordinates method (Viskanta and Mengüç, 1987). The next section addresses this issue.

Implicit Solution for Strongly Participating Medium. For applications that involve strongly participating media ($\kappa^* \geq 4$ for a plane layer), the explicit update of R becomes unsatisfactory. This section describes a method that enhances convergence for large κ^*. The method is, for simplicity, presented in the context of the present one-dimensional problem, but the derivation is kept sufficiently general to include nonisotropic scattering.

To re-introduce nonisotropic scattering, R in equation (19C) is replaced by R^l, and the nodal values of R are split as follows:

$$R_P^l = \frac{K_P I_{b,P} + \sigma_P^s I_{a,P}}{\kappa_P} + \frac{\sigma_P^s}{\kappa_P}(\bar{I}_P - I_{a,P}) = \bar{R}_P + \tilde{R}_P^l \qquad (21)$$

For isotropic scattering, $\tilde{R} = 0$. Expressing the integration point values of R_f^l and $(\partial R^l/\partial z)_f$, $f = s$ and n, in equation (19C) in terms of nodal point values using linear interpolation, substituting the resultant b_P^l expression into equation (19A), and dividing by a_P^l, yields an equation for I_P^l of the form

$$I_P^l = \frac{a_S^l}{a_P^l} I_S^l + \frac{a_N^l}{a_P^l} I_N^l - A_P^l \bar{R}_P + A_S^l \bar{R}_S + A_N^l \bar{R}_N + B_P^l \qquad (22)$$

where terms involving $\tilde{R}^l$ have been grouped into B_P^l. Multiplication of equation (22) by ω^l, summation over all ω^l, and division of the resulting equation by 4π yields an equation of the form

$$I_{a,P} = -\hat{A}_P \bar{R}_P + A_S \bar{R}_S + A_N \bar{R}_N + B_P \qquad (23)$$

For the special case of isotropic scattering and radiative equilibrium, $\bar{R}_P = I_{a,P}$ (from equation (16)) so that equation (23) becomes

$$A_P I_{a,P} = A_S I_{a,S} + A_N I_{a,N} + B_P \qquad (24)$$

which is an implicit equation for $I_{a,P} = I_{b,P}$ that is simply solved by application of the Thomas (or tridiagonal) algorithm. This replaces the "explicit" update of R described in the previous section. It is also easily shown that, as $\kappa^* \to \infty$, equation (24) becomes the discrete representation of the diffusion approximation of radiation heat transfer. For large κ^*, therefore, the first solution for $I_{a,P}$ will yield nearly the correct $I_{a,P} = R_P$ values, which, used in the b_P^l term in equation (19A), will yield nearly converged values of I_P^l, so that very few cycles will lead to the converged result.

Figure 5 shows the number of cycles required by this implicit method to obtain surface heat transfers that changed less than 0.001 percent between R-update cycles. The maximum number of cycles is required for $\kappa^* \approx 20$. For $\kappa^* \to 0$ and $\kappa^* \to \infty$, the converged results are obtained in one solution cycle.

Although the method has not yet been applied to problems involving nonisotropic scattering and radiative nonequilibrium, it is interesting to speculate how the implicit method could be used to accelerate convergence. Splitting $\bar{R}$ in equation (23) into its I_a and I_b components results in an equation of the form

$$A_P^{aa} I_{a,P} = A_S^{aa} I_{a,S} + A_N^{aa} I_{a,N} - A_P^{ab} I_{b,P} + A_N^{ab} I_{b,N} + A_S^{ab} I_{b,S} + B_P^a \qquad (25)$$

where the first superscript on the coefficients refers to the equation and the second to the variable it multiplies (either a for the variable I_a or b for the variable I_b).

The thermal energy equation has diffusion and convection terms that are linear in T, and a source term given by equation (14B) that contains I_a and $I_b = \sigma T^4/\pi$. If radiation dominates the internal energy balance, it is appropriate to linearize the discrete energy equation in the form

$$A_P^{bb} I_{b,P} = A_S^{bb} I_{b,S} + A_N^{bb} I_{b,N} + A_P^{ba} I_{a,P} + B_P^b \qquad (26)$$

Equation (25) and (26) can be efficiently solved for $I_{a,P}$ and $I_{b,P}$ (e.g., use a (2×2) block tridiagonal solver). If radiation only weakly affects the medium temperature, it would be more appropriate to linearize the equations such that T instead of I_b appears as the dependent variable in equations (25) and (26). In either case, iteration is required to converge to the solution of the equations in which some terms involve T and others T^4.

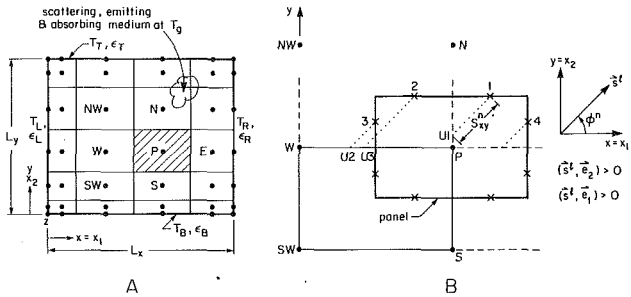


Fig. 6 Schematic of a rectangular cavity with isothermal walls with a Cartesian computational grid. Part B shows details related to determination of the integration point intensities for those intensities that are directed upward and to the right.

Two-Dimensional Exchange in Rectangular Enclosures

Problem Description. The finite volume method will now be applied to a more complex class of problems that involve two-dimensional (2D) radiative heat transfer in rectangular enclosures with isothermal, diffusely reflecting, gray walls. The enclosed medium is also assumed to be gray, and to scatter isotropically. Figure 6(A) shows a typical enclosure. The objective is to predict radiative heat transfer at the boundaries, and the medium temperature under the assumption of radiative equilibrium. To test the finite volume method, several benchmark problems will be solved. To simplify notation in the development that follows, the (x_1, x_2, x_3) coordinates will be used interchangeably with (x, y, z).

Discretization. Space is divided into control volumes of unit depth in the z direction and with boundaries defined by the solid lines in Fig. 6(A). Nodes, also called grid points, are located centrally in each interior volume as well as on the boundary. Each control volume face is divided into two panels with an integration point, such as 1, 2, 3, in Fig. 6(B), located at the center of each panel.

The angular orientation of intensity is defined by a polar angle θ, measured from the $x_3 = z$ axis, and the azimuthal angle ϕ, that lies in the (x, y) plane measured from the x axis. The solid angle ω^l is defined by the range $\theta_-^m \leq \theta \leq \theta_+^m$, $\phi_-^n \leq \phi \leq \phi_+^n$. All results shown are for N^θ equal increments of θ in the range $0 \leq \theta \leq \pi/2$ and for N^ϕ equal increments in ϕ in the range $0 \leq \phi \leq 2\pi$, where, for simplicity, N^ϕ was always chosen to be a multiple of 4.

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To apply the general equations of section 2 to the grid system just described, some details must be decided. The central ray within ω^l, i.e., the one along s^l, is defined by (θ^m, ϕ^n), where each angle lies in the middle of its range (e.g., $\phi^n = (\phi^n_+ + \phi^n_-)/2$). The path length $\bar{S}^l$ is taken as the distance from the integration point to the point where the $-\mathbf{s}_l$ that passes through the integration point intersects the closest member of the set of surface that join the grid points. For example, in Fig. 6(B), for integration point $f = 1$

$$\bar{S}^l = S^n_{xy}/\sin\theta^m$$

where S^n_{xy} is the projection of $\bar{S}^l$ onto the x–y plane.

Intensities at u_1, u_2, etc. in Fig. 6(B) must be related to the nodal intensities by interpolation functions. This step requires great care. I^l_{u1} could, for example, be found by linear interpolation between I^l_N and I^l_P, but this would cause I^l_N to appear in the I^l_P equation. This is undesirable because the equation for I^l_P can no longer be solved by marching, and because such dependence of "downstream" intensities is inconsistent with the transportive properties of the differential equation for intensity. A better alternative is to find I_{u1} by extrapolation using I^l_P and I^l_S, but applying this practice at each integration point results in a computational molecule that connects I^l_P with as many as six upstream nodal intensities. While this is fairly easily accommodated, a simpler approach was used for the present application. When (for s^l upward and to the right) an interpolation point lies on a solid line in Fig. 6(B), linear interpolation between nodes was used. When it lies on a broken line, the interpolation value was taken as the value at the nearest "upstream" node on the line. According to this scheme I^l_{u1} and I^l_{u4} would be both approximated by I^l_P for $(\mathbf{s}^l \cdot \mathbf{e}_j) \geq 0, j = 1, 2$. I^l_{u2} and I^k_{u3} would both be found by a linear interpolation involving I^l_P and I^l_W.

If linear interpolation or extrapolation were used to find all interpolation-point values, the solution error should decrease quadratically with mesh refinement, as for the one-dimensional problem, but it is well known (from experience in approximating fluid advection) that "wiggles" in the solution can occur. The interpolation practice adopted here is less likely to admit wiggles, but it is expected that errors due to spatial discretization will decrease with grid refinement at a rate that is between a linear and quadratic. These are features that will be examined when the benchmark problems are solved.

Solution Procedure. For intensities in those $\mathbf{s}^l$ directions that are upward to the right, the discrete equation for I^l_P has the form

$$a^l_P I^l_P = a^l_W I^l_W + a^l_S I^l_S + a^l_{SW} I^l_{SW} + b^l_P \qquad (27)$$

Given the boundary values of I^l along the left and bottom surfaces in Fig. 6(A), nodal values of I^l_P are found by direct substitution into equation (27), sweeping the domain in the positive x and y directions. With new incident intensities on the right boundary known, boundary conditions are applied to update the intensities leaving this surface, and an equation similar to equation (27) establishes nodal values for $\mathbf{s}^l$ in the range $(\mathbf{s}\cdot\mathbf{e}_1) < 0$ and $(\mathbf{s}\cdot\mathbf{e}_2) > 0$. Repeating this for all four sectors leads to a new I^l_P solution, for a given R_P distribution. R_P is then explicitly updated by calculating $I_{aP} = R_P$ from equation (13A).

For simplicity, the coefficients of equation (27) were recalculated each time the set of intensities associated with a particular ω^l was updated. The run times for the problems solved could be dramatically reducted by storing coefficients or even by using simple, computationally inexpensive approximations to the exponential terms. The intensities were stored in a four dimensional matrix, also for simplicity. For isotropic scattering, it is only necessary to store the complete I^l_P values at boundary nodes so that four-dimensional storage is not required.

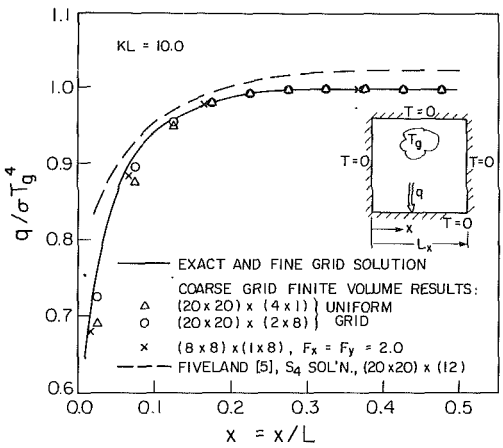


Fig. 7 Nondimensional heat transfer on the wall of a square cavity containing an emitting gas at uniform temperature T_g and with zero wall temperature for an optically thick case ($KL = 10.0$)

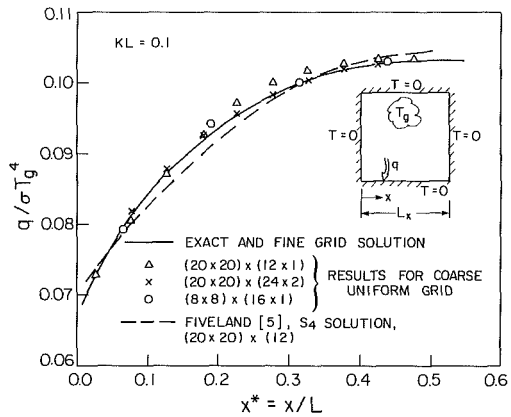


Fig. 8 Nondimensional heat transfer on the wall of a square cavity containing an emitting gas at uniform temperature T_g and with zero wall temperature for an optically thin case ($KL = 0.1$)

Comparison With One-Dimensional Solutions. The two-dimensional solution was applied first to the one-dimensional problem described in the last section, for a range of surface emissivities. As the grid was refined, the two-dimensional model converged to the same result as the one-dimensional model.

Problem 1: Black Enclosure With Emitting Medium. The first benchmark problem is the prediction of surface heat transfer in a square enclosure ($L_x = L_y = L$) with cold walls (at 0 K), which contains a gas at uniform temperature, T_g. The gas does not scatter ($\sigma^s = 0$) but does emit and absorb ($K \neq 0$). With the gas temperature known, the value of R in equation (10) is prescribed.

The local heat flux calculated to fall on the lower wall is plotted in Figs. 7 and 8 for the cases of strongly emitting medium ($KL = 10$) and weakly emitting medium ($KL = 0.1$), respectively. The "exact" solution was obtained by analytically integrating the radiation transfer equation for a given angle to the point of interest on the wall from the "upstream" wall, and then numerically integrating over all angles to find the local heat transfer. The present finite volume model could be made to agree with these exact results to any desired tolerance by refining the spatial and angular grids.

The S_4 discrete ordinates solution by Fiveland (1984) to this problem, using a (20×20) uniform spatial grid and with twelve quadrature points for the angular integrations are shown on Fig. 7 for $KL = 10$. Truelove (1987) was able to improve Fiveland's discrete ordinates solution by changing the

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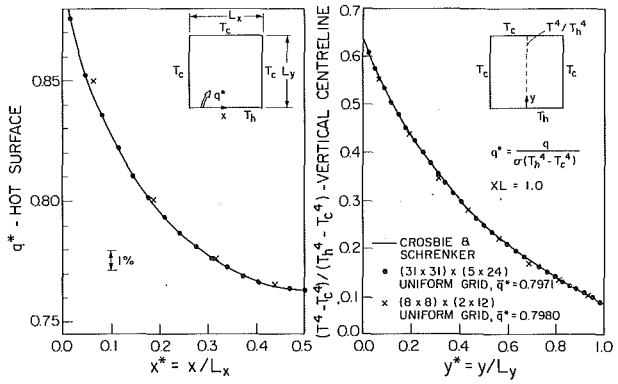


Fig. 9 Nondimensional heat transfer on the bottom, and centerline temperature distribution, in a square cavity with one heated wall enclosing a scattering gas with $\sigma^s L = 1.0$

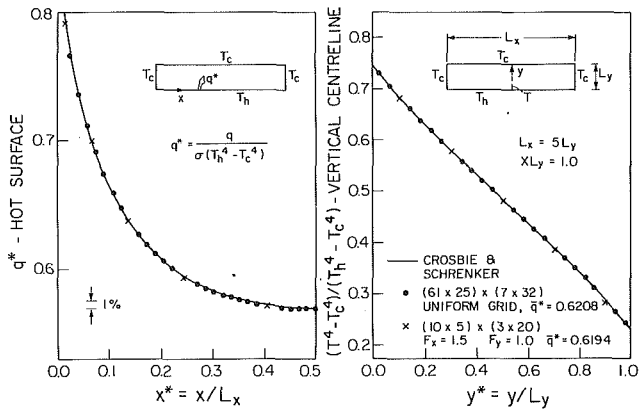


Fig. 11 Nondimensional heat transfer on the bottom, and centerline temperature distribution, in a 5:1 aspect ratio cavity enclosing a scattering gas with $\sigma^s L_y = 1.0$

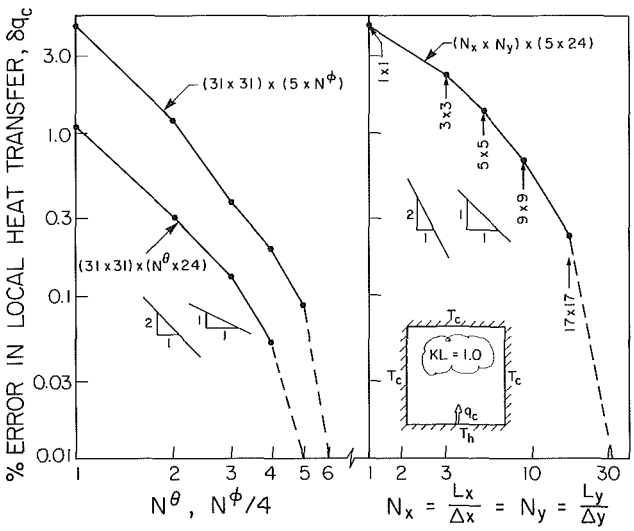


Fig. 10 Reduction in the error in local heat transfer at the center of the heated cavity wall with angle refinement (left side) and spatial refinement (right side)

quadrature points. Comparisons are made here with Fivelands solution, because the quadrature he used was not tuned to this particular class of problem. The present finite volume solutions for 20 uniform control volumes in each direction, $(N_x \times N_y) = (20 \times 20)$, and with a uniform angular discretization of $(N^\theta \times N^\phi) = (1 \times 4)$ and (2×8) are also shown in Fig. 7, and they give reasonable agreement with the exact solution. Reducing the spatial grid to (8×8) but using an expanding mesh that doubles the size of each successive control volume moving from the corners to the centerlines ($F_x = F_y = 2.0$ in Fig. 7), and using a uniform (1×8) angular discretization, results in excellent agreement with the exact solution. The solution time for $(20 \times 20) \times (2 \times 8)$ problem was 15 s on a MicroVax II computer.

Similar results appear in Fig. 8 for the $KL = 0.1$ case. The finite volume results obtained using all the coarse grids shown in Figs. 7 and 8 predict a total heat transfer on the bottom, which agrees with the exact values to within 0.8 percent.

The results for this sample problem illustrate that the finite volume method converges to the exact solution as the angular and spatial grids are refined, and that predictions of good accuracy can be obtained on quite coarse grids.

Problem 2: Two-Dimensional Heat Transfer in an Enclosure With One Hot Wall. The second two-dimensional problem involves radiative exchange in rectangular enclosures with the left, top, and right wall temperatures set to T_c and the bottom wall set a higher temperature T_h. The medium scatters isotropically and is assumed to be in radiative equilibrium. The objective is to predict the heat transfer that leaves the bottom surface as well as the temperature of the medium. Careful solutions to this problem have been reported by Crosbie and Schrenker (1984) for a range of cavity aspect ratios and scattering coefficients. Comparisons with finite volume predictions are presented here for three cases.

The "exact" (Crosbie and Schrenker, 1984) dimensionless local heat transfer q^* leaving the hot wall and temperature distribution along a vertical line in the center of a square enclosure are plotted as solid lines in Fig. 9 for a medium with $\sigma^s L = 1.0$. Application of the finite volume method with a very fine mesh $(N_x \times N_y) \times (N^\theta \times N^\phi) = (31 \times 31) \times (5 \times 24)$ is shown to give almost exact agreement (local heat fluxes agree to well within 0.1 percent). Use of a coarse $(8 \times 8) \times (2 \times 12)$ grid, such as would be of more practical interest, also yields excellent accuracy, with an average heat transfer $\bar{q}^*$ that is only 0.1 percent too high. To obtain a solution on this grid, in which the absolute value of the heat transfer on all walls changed by less than 0.05 percent between R-updated cycles, required eight cycles and 32 s on a MicroVax II computer.

As previously discussed, it is important to know how quickly the solution error decreases with grid refinement. To perform this evaluation, the error in the local heat transfer at the center of the hot wall ($x/L = 0.5$) was examined. Retaining (32×32) uniform control volumes, fixing N^ϕ at 24, and varying N^θ, Fig. 10 shows that the error diminishes quadratically with N^θ. Similarly, for N^θ fixed at 5, increasing N^ϕ results in a quadratic decrease in error. For a fine angular grid (5×24) and with refinement of a uniform spatial grid, the error is seen in Fig. 10 to decrease somewhere between linear and quadratic with increasing $N_x = N_y$. These results are in agreement with the expectations that were previously discussed.

For an enclosure with an aspect ratio of $L_x/L_y = 5$, the "exact solution" of Crosbie and Schrenker (1984) and a fine-grid $(61 \times 25) \times (7 \times 32)$ finite volume solution are plotted in Fig. 11 for $\sigma^s L_y = 1.0$. The finite volume predictions of heat flux agree with the exact values to within 0.2 percent, and the temperature is also closely predicted. Figure 11 also shows that a much coarser grid $(10 \times 5) \times (3 \times 20)$, where the control volumes widths in the x direction expand by a factor of 1.5 moving from the outer walls toward the center ($F_x = 1.5$), yields virtually identical results, with an average heat transfer $(\bar{q}^*)$ just 0.2 percent below the fine grid prediction. The prediction on the $(10 \times 5) \times (3 \times 20)$ grid required five R-update cycles and 17.6 s to achieve a total absolute value of heat transfer on the boundaries that changed less than 0.05 percent between cycles.

The last test problem described in this paper has been specifically chosen to demonstrate that, as mentioned earlier,

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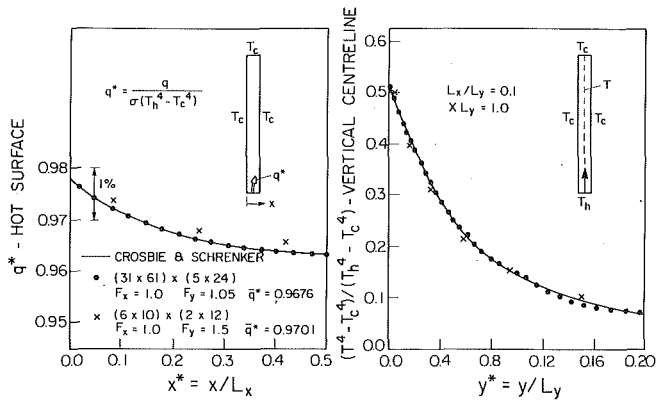


Fig. 12 Nondimensional heat transfer on the bottom, and centerline temperature distribution, in a 1:10 aspect ratio cavity enclosing a scattering gas with $\sigma^S L_y = 1.0$

profile wiggles can occur under adverse conditions. Results for a cavity with $\sigma^s L_y = 1.0$ and $L_y/L_x = 10$ are compared to the predictions of Crosbie and Schrenker in Fig. 12. Calculations on a fine $(31\times61)\times(5\times24)$ grid, with uniform control volumes in the x direction and with volumes in the y direction that increase by expansion factor of $F_y = 1.05$, yielded local heat transfer predictions that are in excellent agreement with Crosbie and Schrenker (Fig. 12). The predicted centerline temperature near the bottom wall (right side of Fig. 12) is also very accurate but, when the aspect ratio of the control volumes become large in the center to top of the cavity, the predicted temperatures oscillate (or wiggle) slightly about the profile of Crosbie and Schrenker. This "wiggle" is a consequence of the assumed linear profile of I^l between nodes in Fig. 6(B), and is analogous to the wiggles that are often tolerated when a linear profile between nodes is used in the approximation of the advective transport terms in fluid flows. Such wiggles can be overcome (Roache, 1972) at the expense of accuracy by adopting a simple "upwind" approximation, or by using a more elaborate discretization.

Figure 12 also shows predictions using a coarse $(6\times10)\times(2\times12)$ mesh that has uniform spacing in the x direction and expands rapidly ($F_y = 1.5$) in the y direction away from the heated wall. This grid yields good temperature predictions, and heat transfer rates that are within 0.3 percent of exact values.

Summary and Discussion

A new finite volume method has been proposed for the prediction of radiative heat transfer that can be implemented on nonorthogonal grids of the type employed to predict complex fluid flows. The theory is advanced for an absorbing-emitting and scattering gray medium in an enclosure with gray diffuse walls, but extensions to a broad bandwidth model and to include walls with other reflective properties are straightforward. For strongly participating media, the method captures the diffusion approximation to the radiant heat transfer equations. The method ensures that the intensity in a given direction, as well as the net radiative heat transfer, satisfy the global conservation constraints. The approximations of the solid-angle integrals also prevent any occurrence of the "ray effects" encountered in discrete ordinate methods (Viskanta, 1987). The method also appears to be the first that shares the philosophy of, and can use the same nonorthogonal boundary fitted mesh as, modern methods for computing fluid flows and that also converges with grid refinement to the exact solution of the radiation transport equation.

The authors are indebted to anonymous reviewers for pointing out that it may be undesirable for a radiation grid to be congruent with the fluid flow grid since the need for grid refinement may occur in different regions. Such situations are already encountered when the internal energy equation is solved over extensive regions of the flow in which the temperature is nearly constant, but in these regions a fine grid is needed to compute the fluid flow. It is common practice in such cases to use the same grid for both the fluid flow and heat transfer for the sake of simplicity. Similarly, for fluid flow and radiation different grids could certainly be used, but at the expense of complexity. Even if different grids were used, it is still desirable to use a philosophically consistent method.

The method has been demonstrated for a one-dimensional problem, and several simple two-dimensional problems, for which benchmark solutions and other approximate solutions are available. The method is shown to yield good accuracy on coarse grids, and errors due to the discrete representations of the continuous equations reduce rapidly with grid refinement.

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